

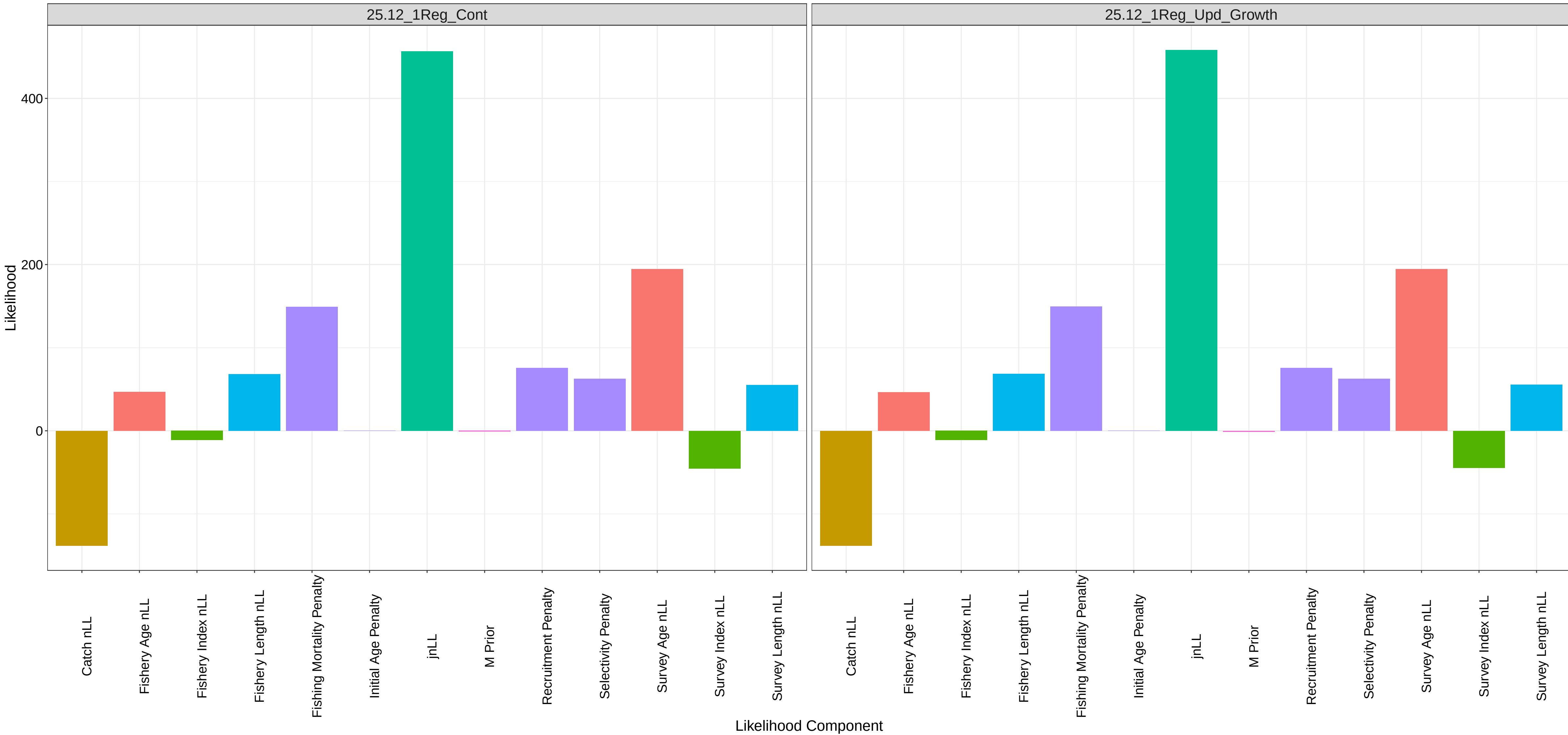
Model		Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
1	25.12_1Reg_Upd_Growth	1	206.7543	358.7654	0.04874091	39.08014	131.7595	0.07902	1.56918	0.57629	0.61680
2	25.12_1Reg_Cont	1	206.6999	357.0208	0.04784668	39.55138	131.4549	0.07875	1.57240	0.57896	0.60758

	Model	jnLL	M Prior	Recruitment Penalty	Selectivity Penalty	Catch nLL	Fishing Mortality Penalty	Survey Index nLL	Fishery Index nLL	Initial Age Penalty	Survey Age nLL	Fishery Age nLL	Survey Length nLL	Fishery Length nLL
1	25.12_1Reg_Upd_Growth	458.1853	-1.103250	75.77622	62.61503	-138.2817	149.5501	-44.65534	-11.10319	0.2654978	194.4674	46.39502	55.49333	68.76618
2	25.12_1Reg_Cont	456.8247	-1.018978	75.86092	62.56423	-138.3008	149.3833	-45.39642	-11.21223	0.2741517	194.6365	46.66608	55.19863	68.16928

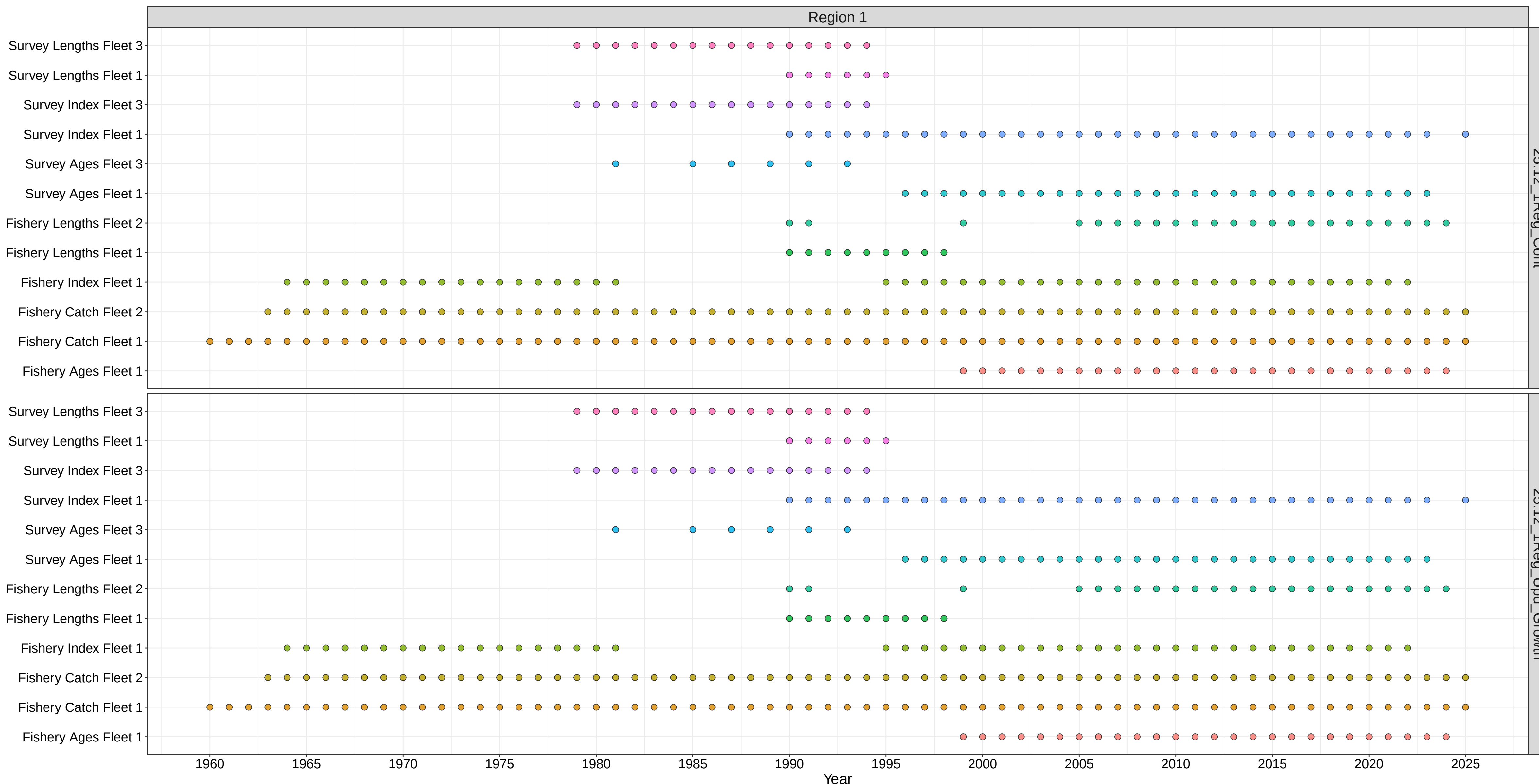
Negative Log-Likelihood

Type

- Age
- Catch
- Index
- jnLL
- Length
- Penalty
- Prior

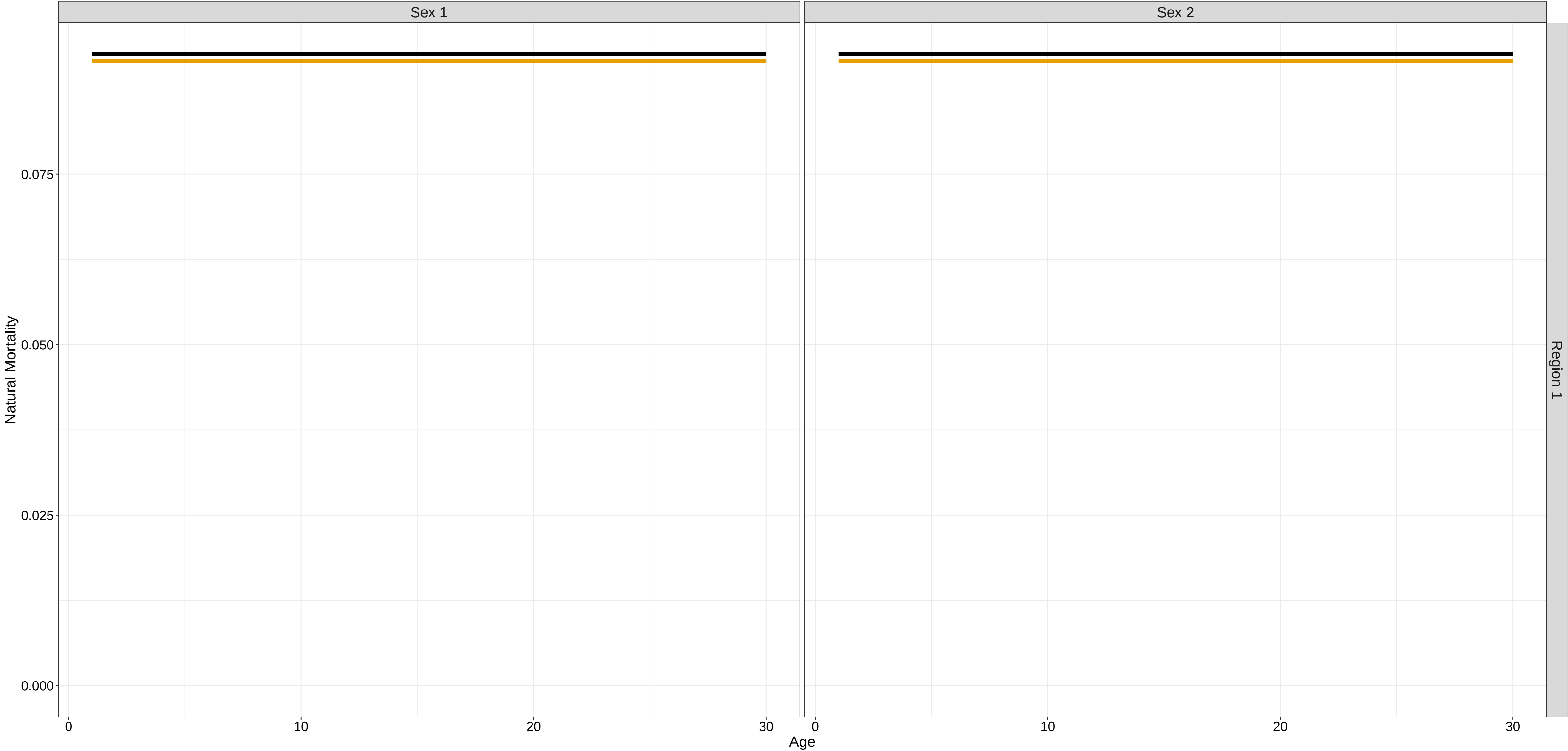


Data



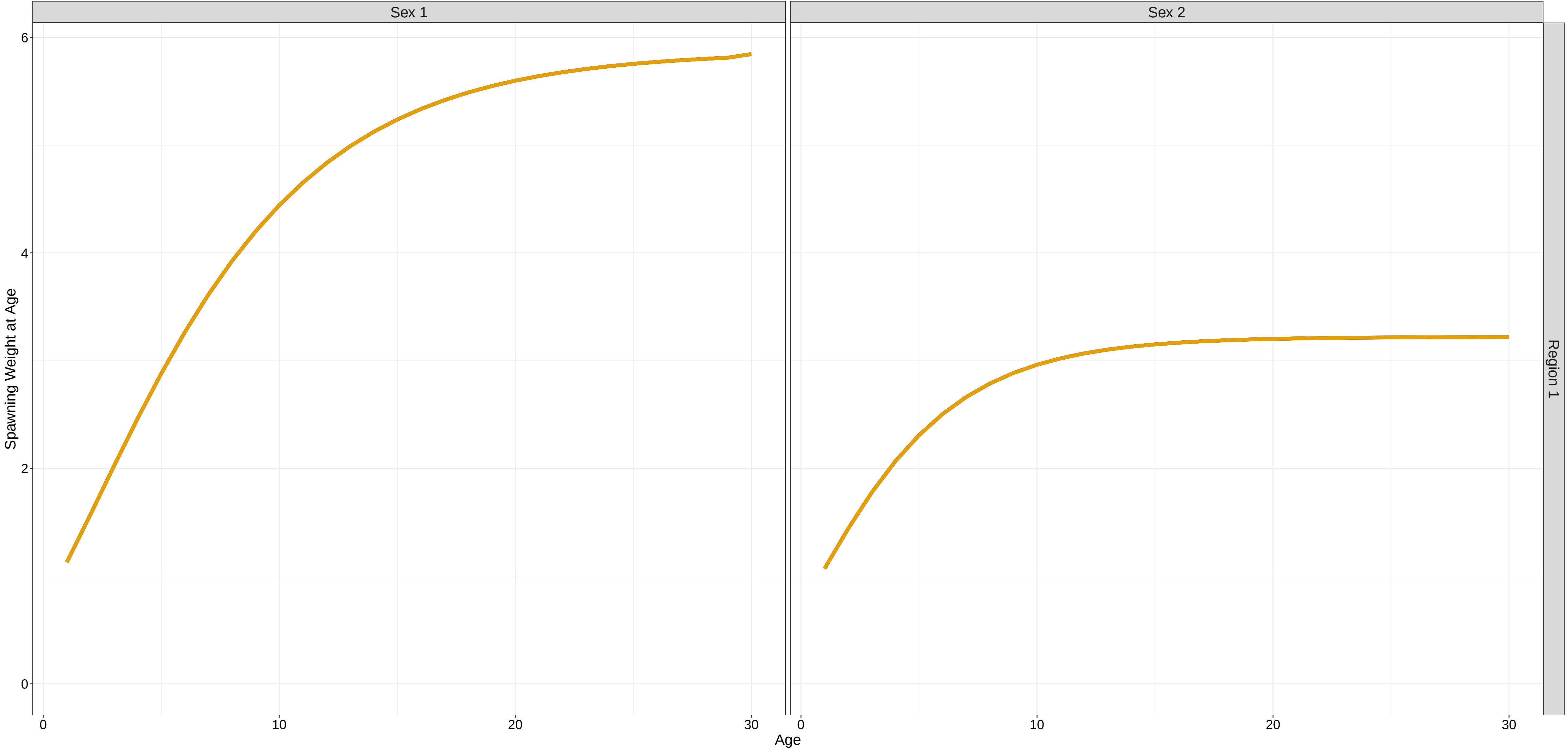
Natural Mortality

Model 25.12_1Reg_Cont 25.12_1Reg_Upd_Growth



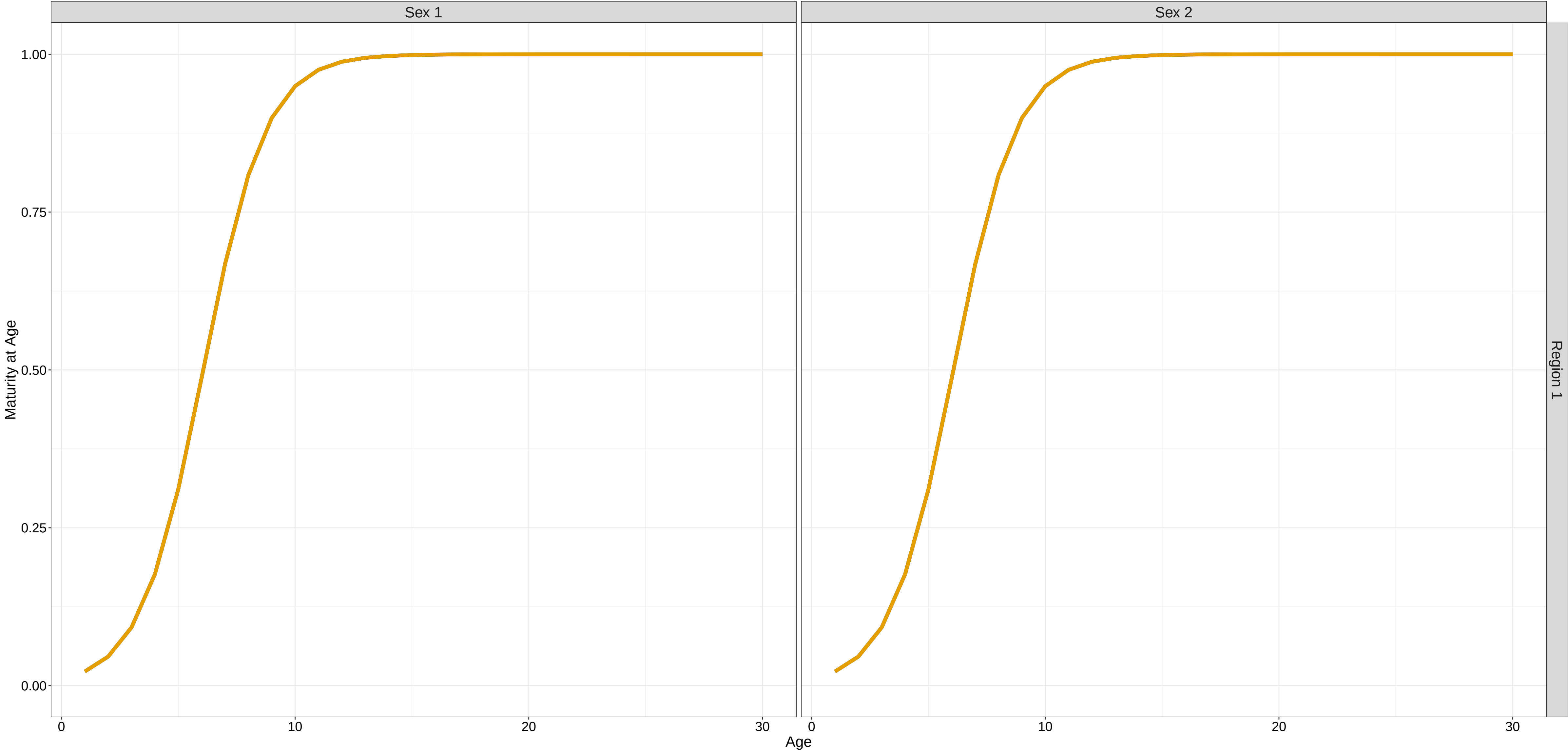
Weight-at-Age

Model 25.12_1Reg_Cont 25.12_1Reg_Upd_Growth



Maturity

Model 25.12_1Reg_Cont 25.12_1Reg_Upd_Growth

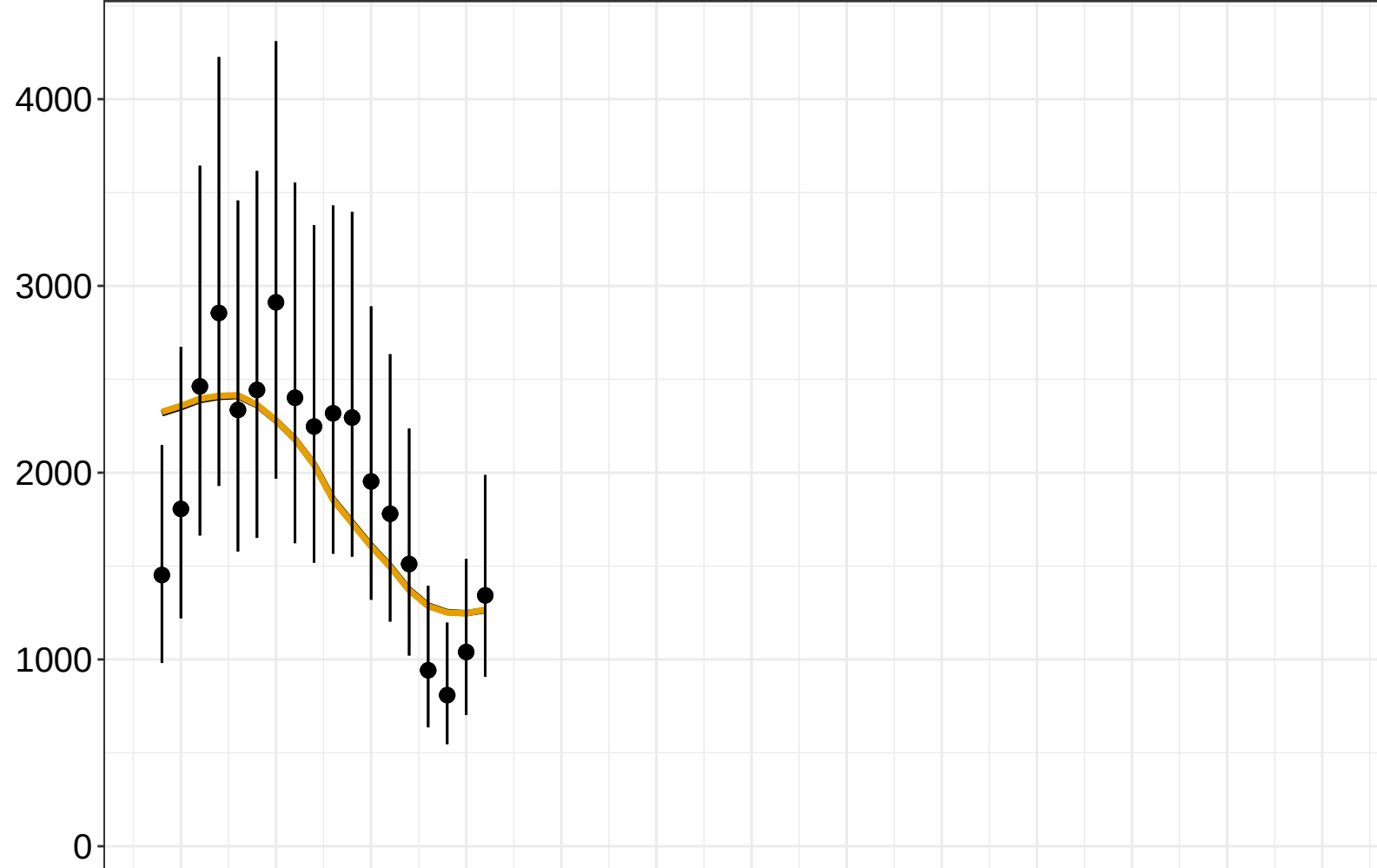


Index Fits

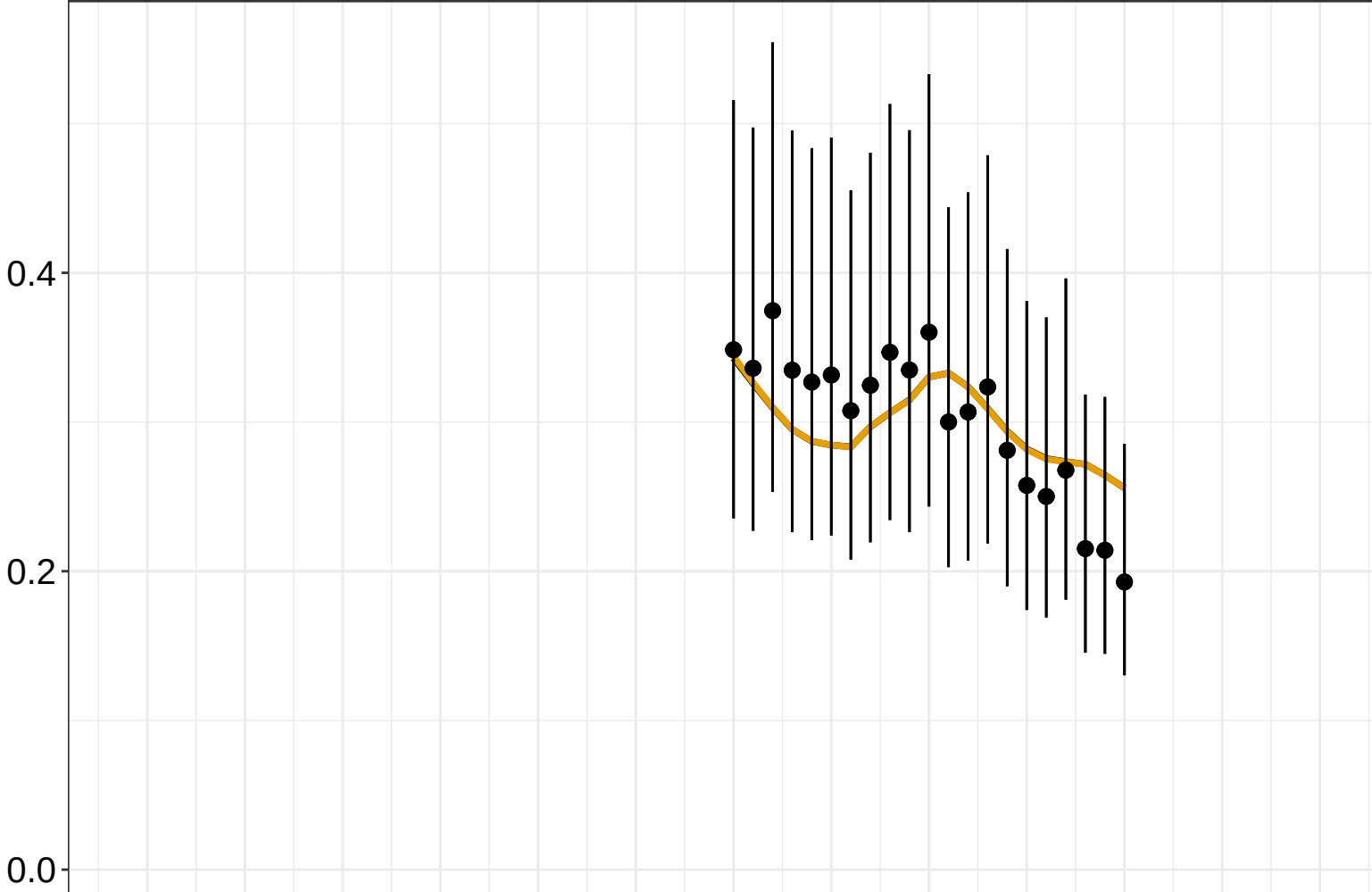
Model — 25.12_1Reg_Cont — 25.12_1Reg_Upd_Growth

Index

JPN Fishery CPUE



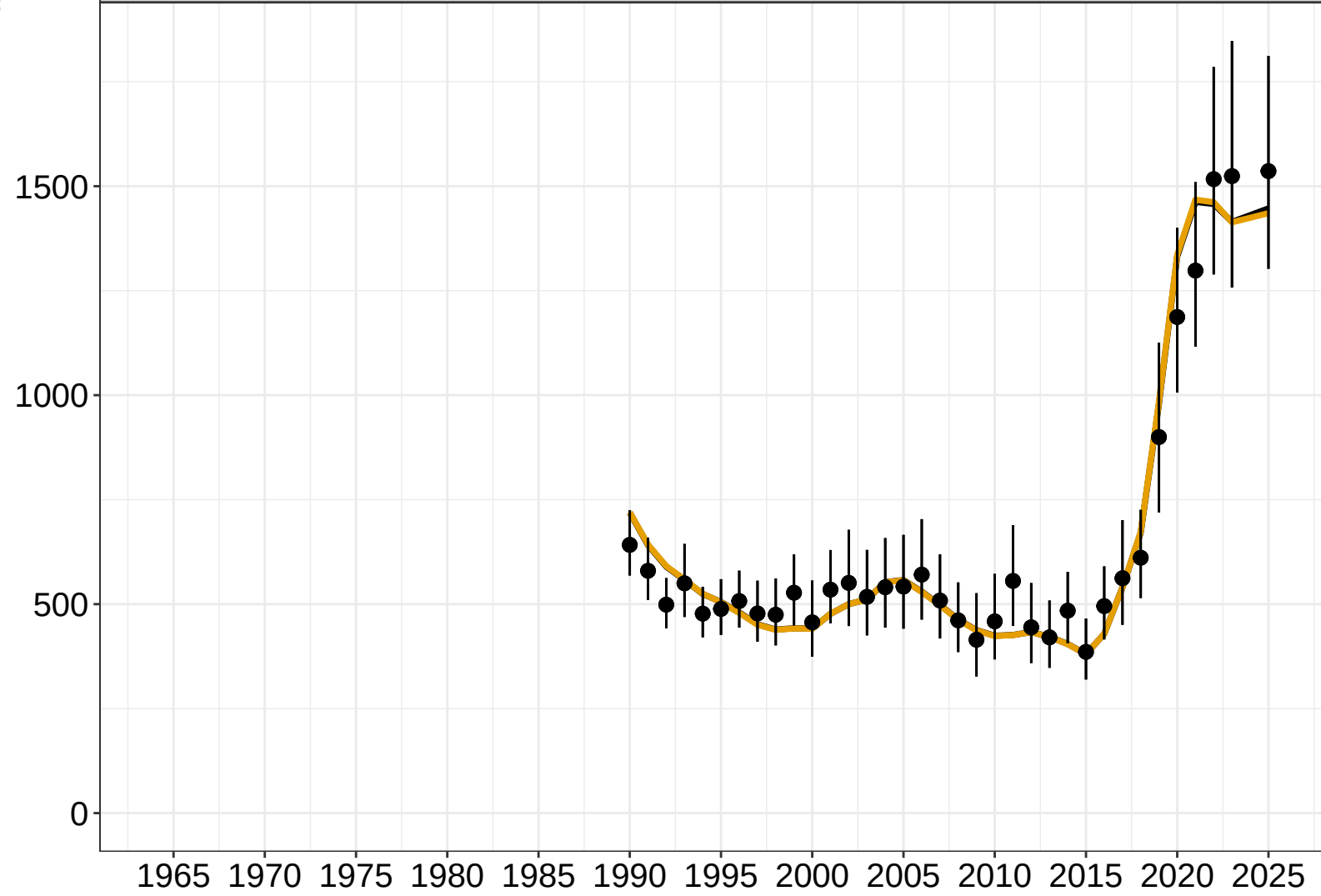
Fishery CPUE



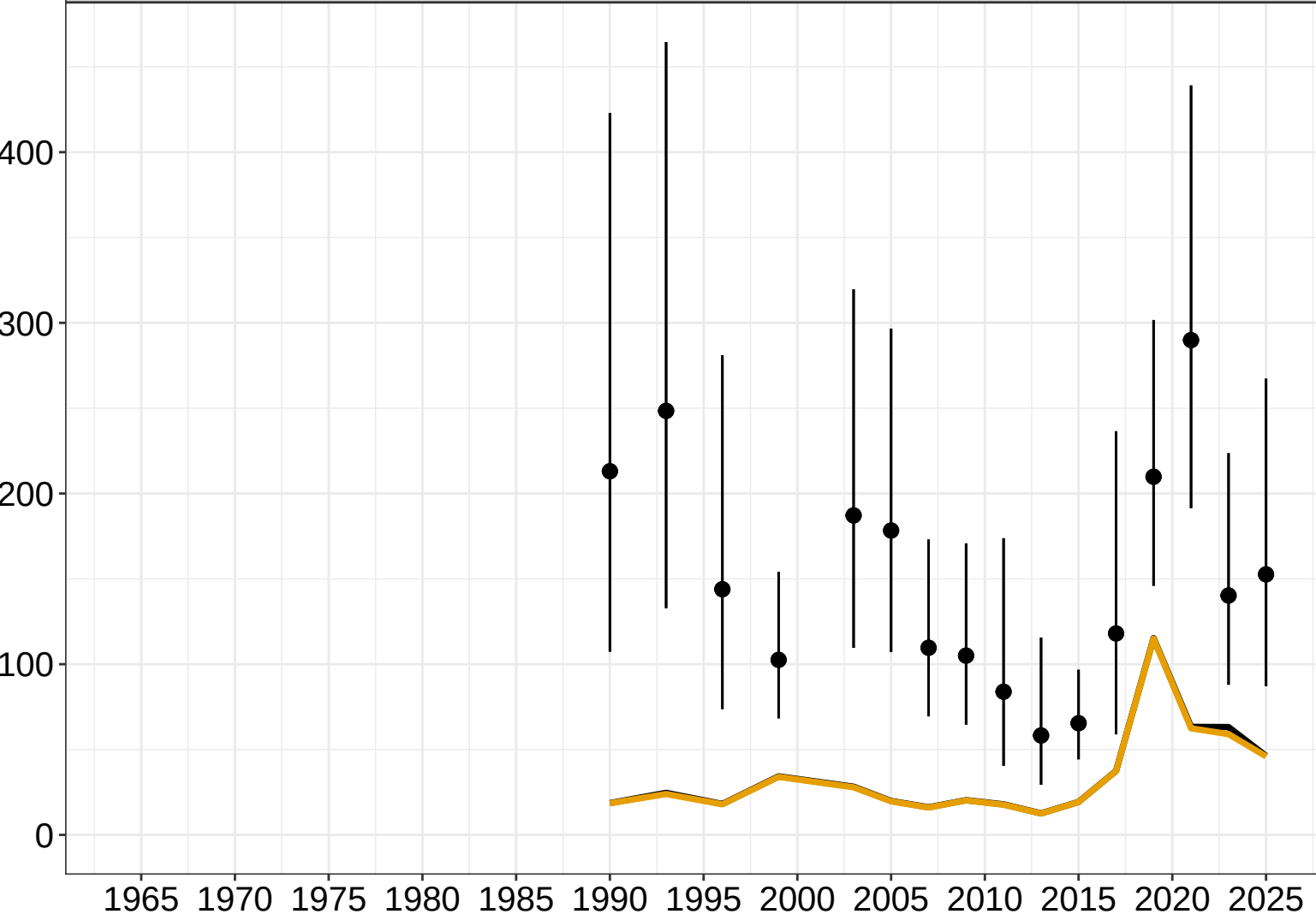
Fishery CPUE



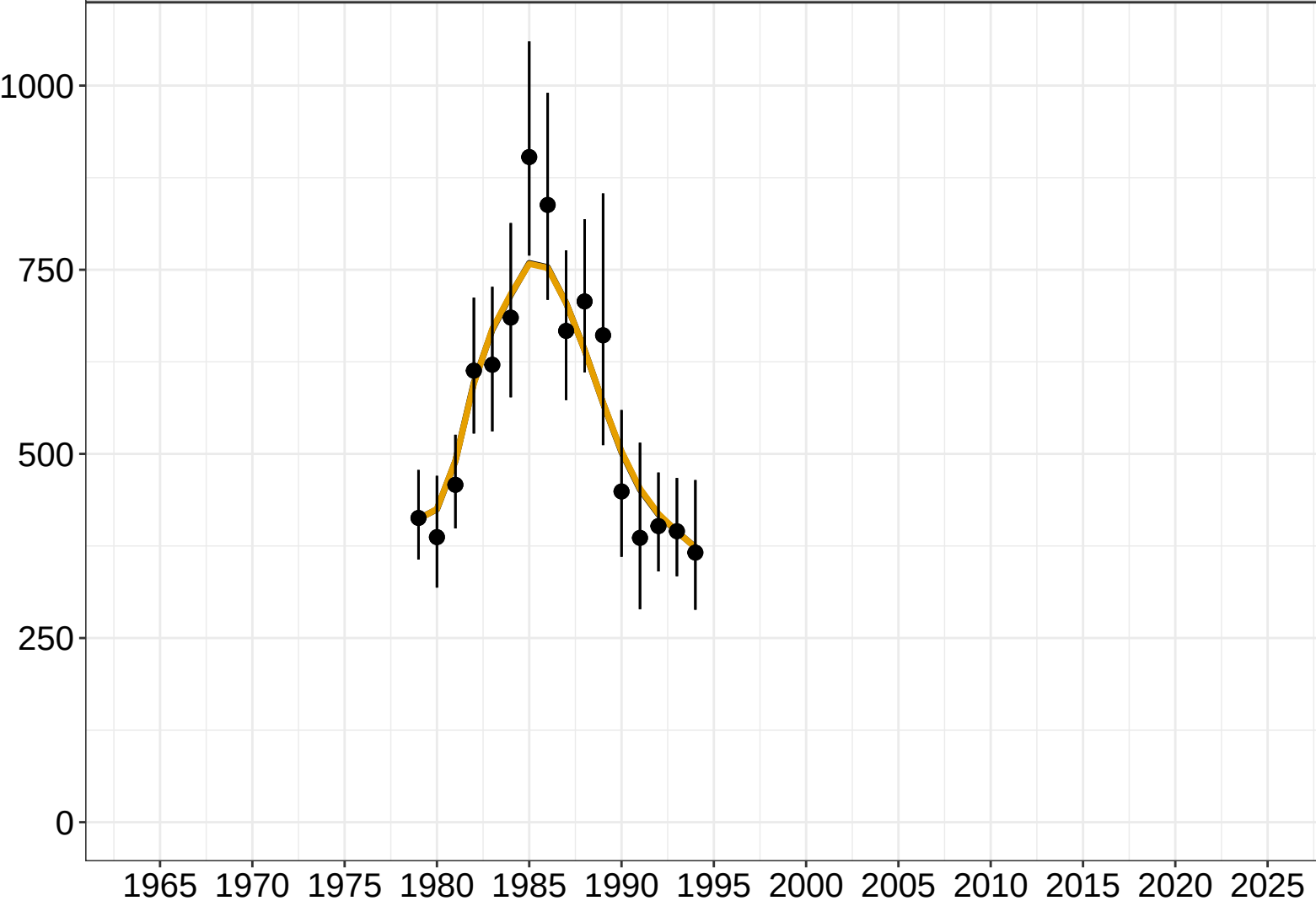
LL Survey RPNs



GOA Trawl Survey (kt)



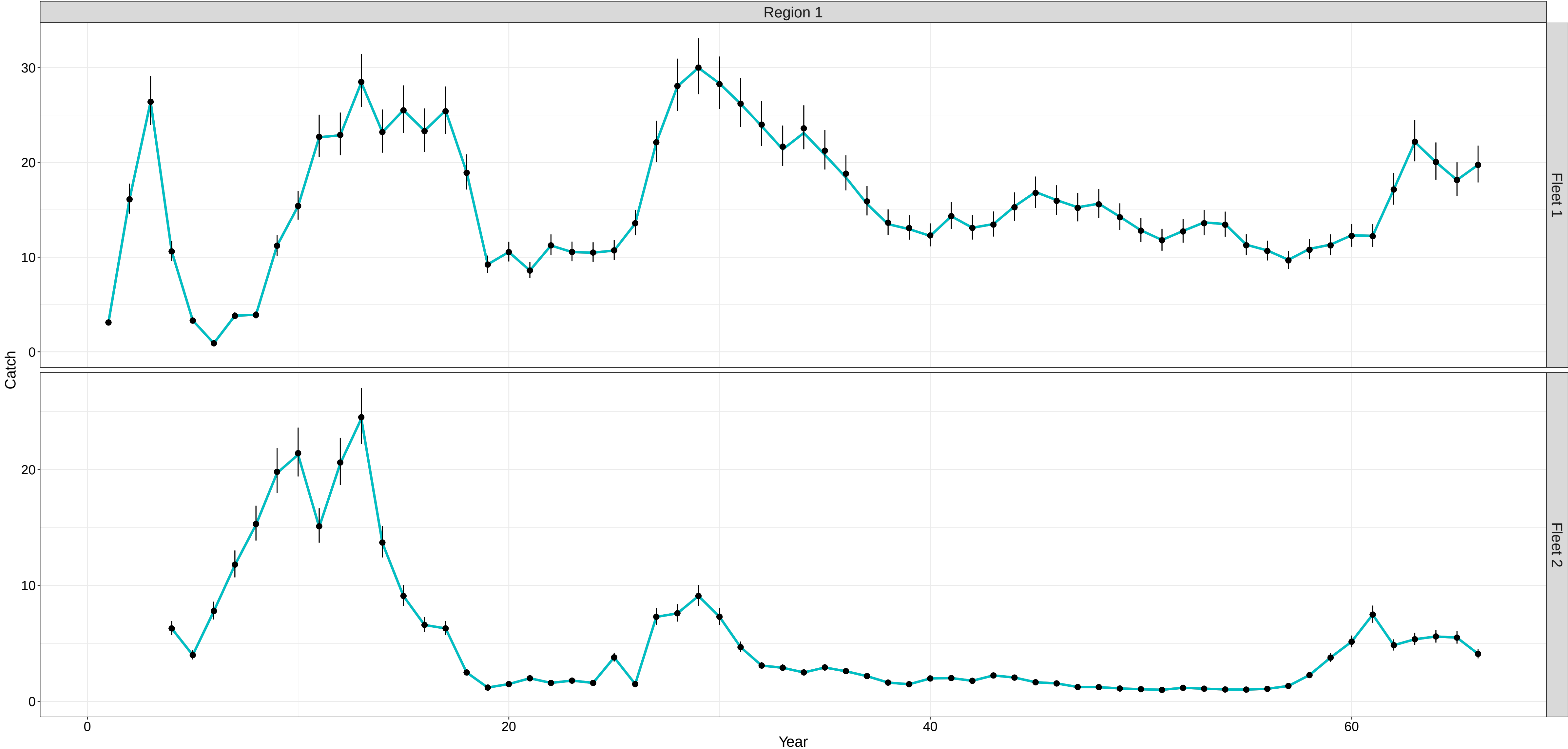
JPN LL Survey RPNs



Year

Catch

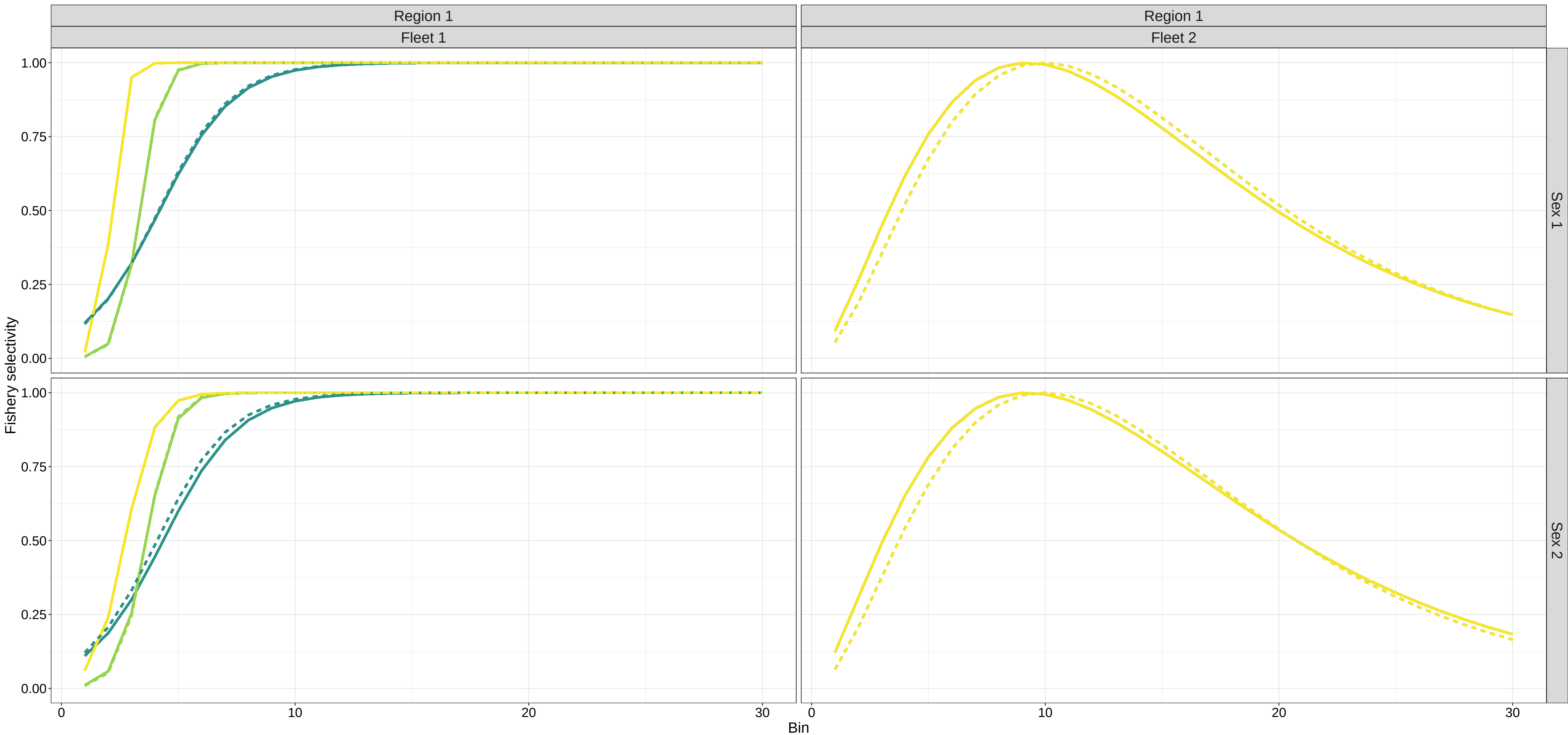
Model 25.12_1Reg_Cont 25.12_1Reg_Upd_Growth



Fishery Selectivity

Model — 25.12_1Reg_Cont - - - 25.12_1Reg_Upd_Growth

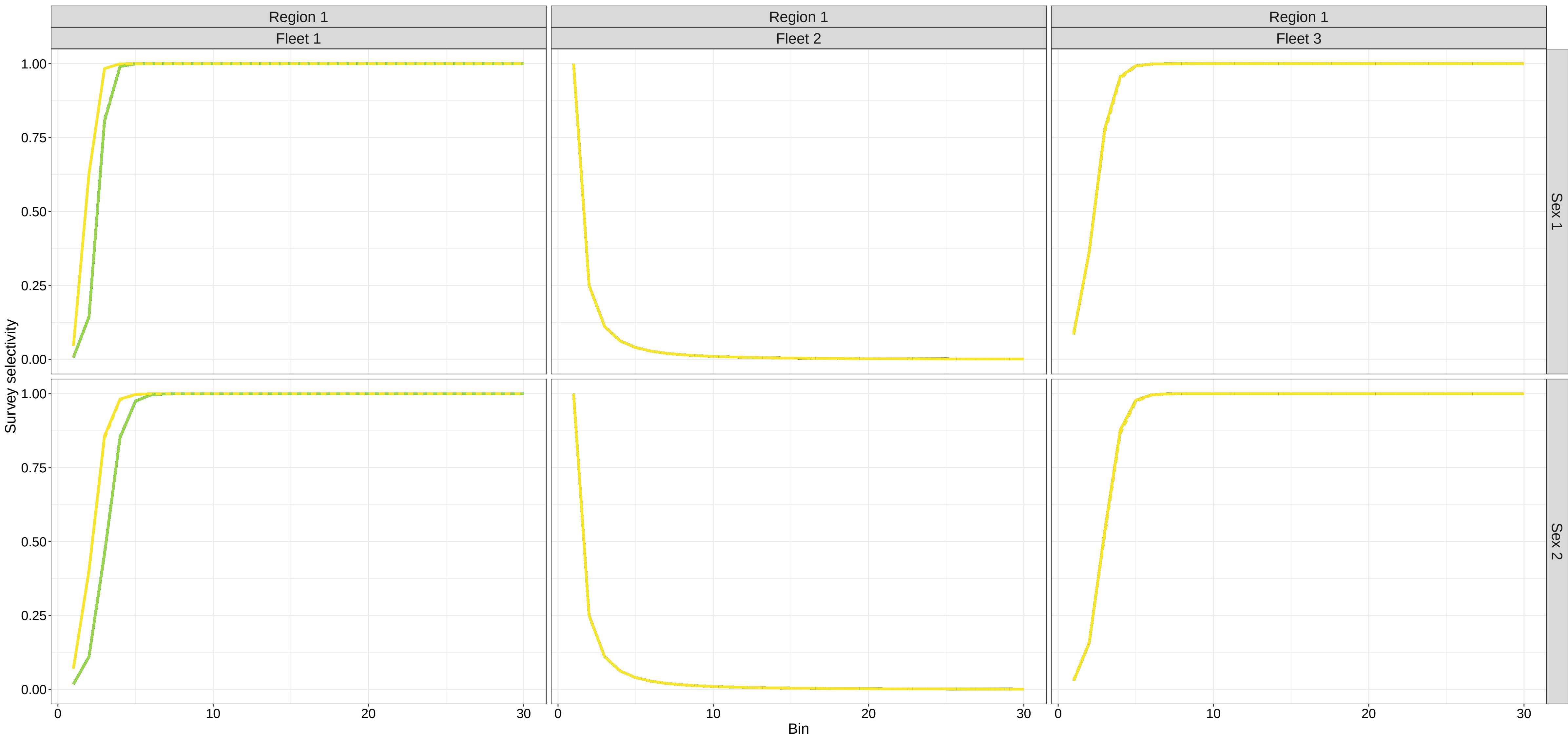
Year  20 40 60



Survey Selectivity

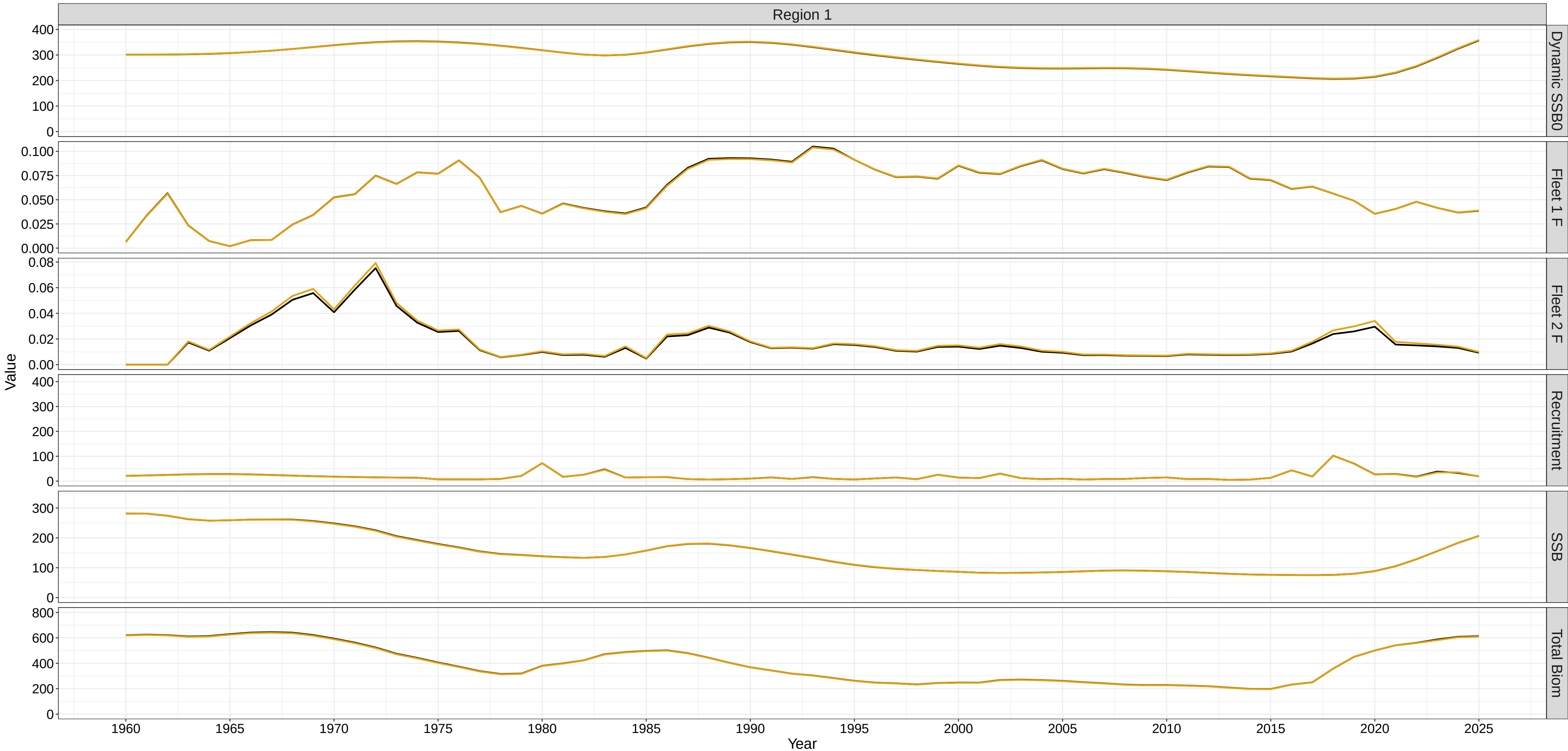
Model — 25.12_1Reg_Cont - - - 25.12_1Reg_Upd_Growth

Year  20 40 60



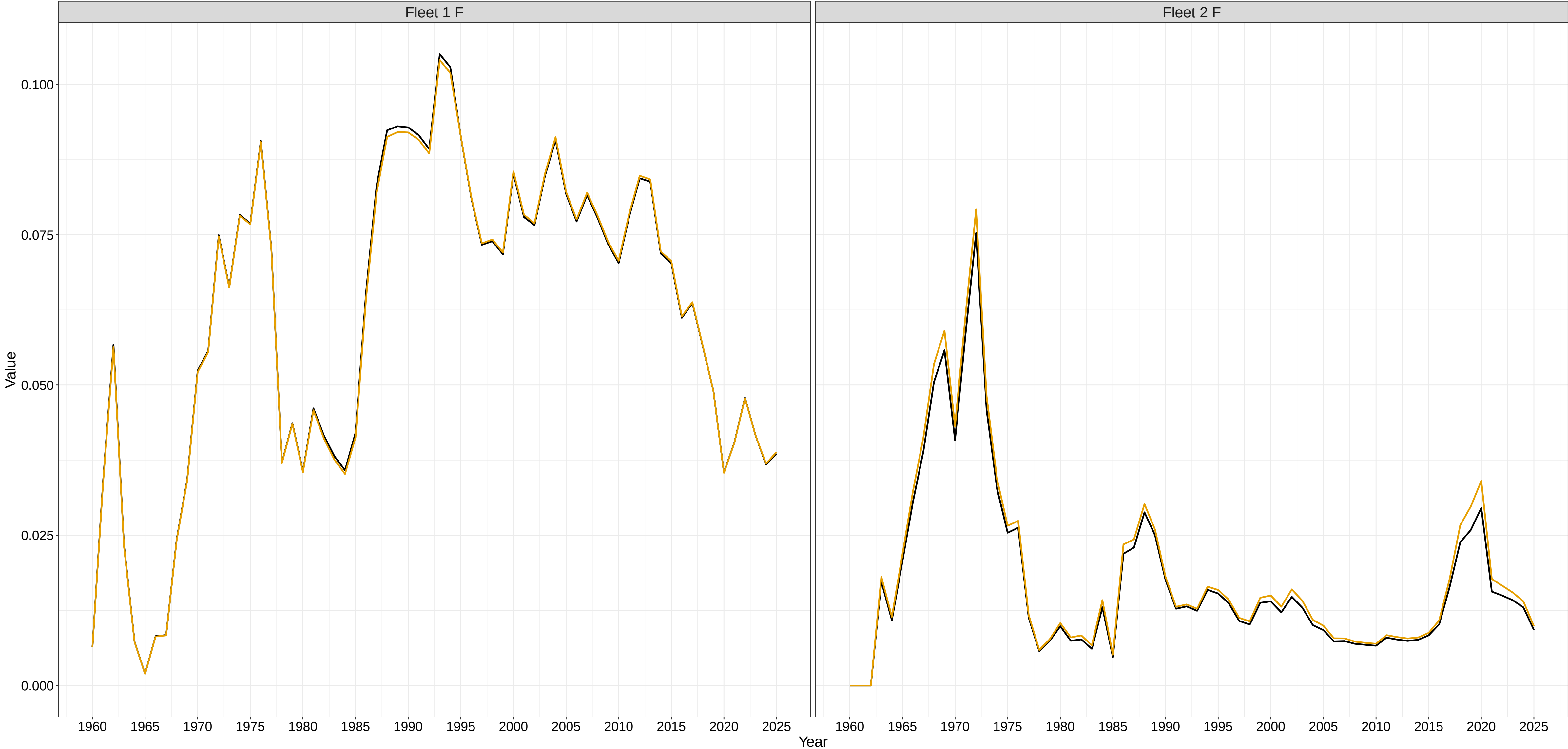
Time Series Plots

Model — 25.12_1Reg_Cont — 25.12_1Reg_Upd_Growth

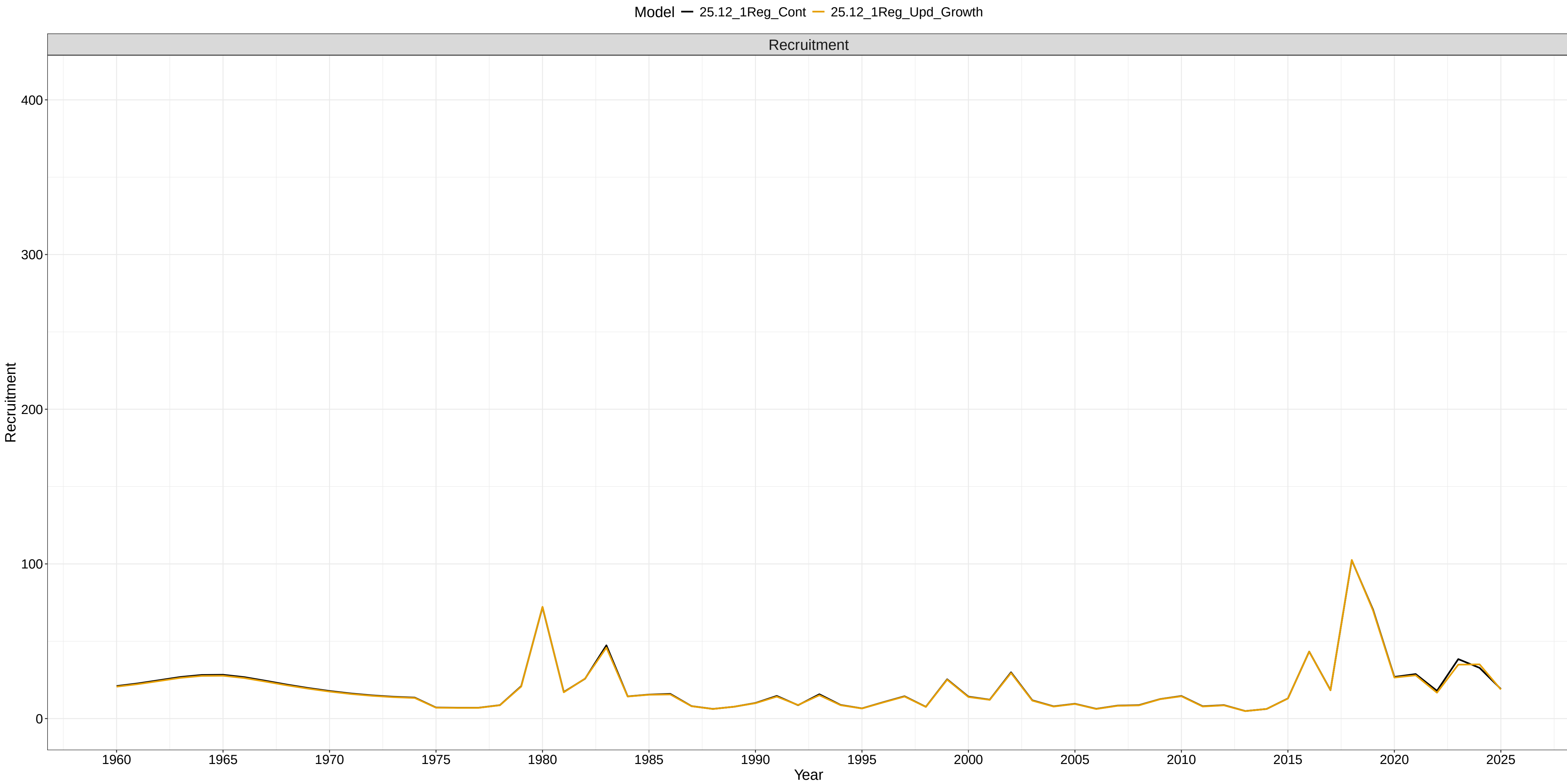


Fishing Mortality

Model — 25.12_1Reg_Cont 25.12_1Reg_Upd_Growth

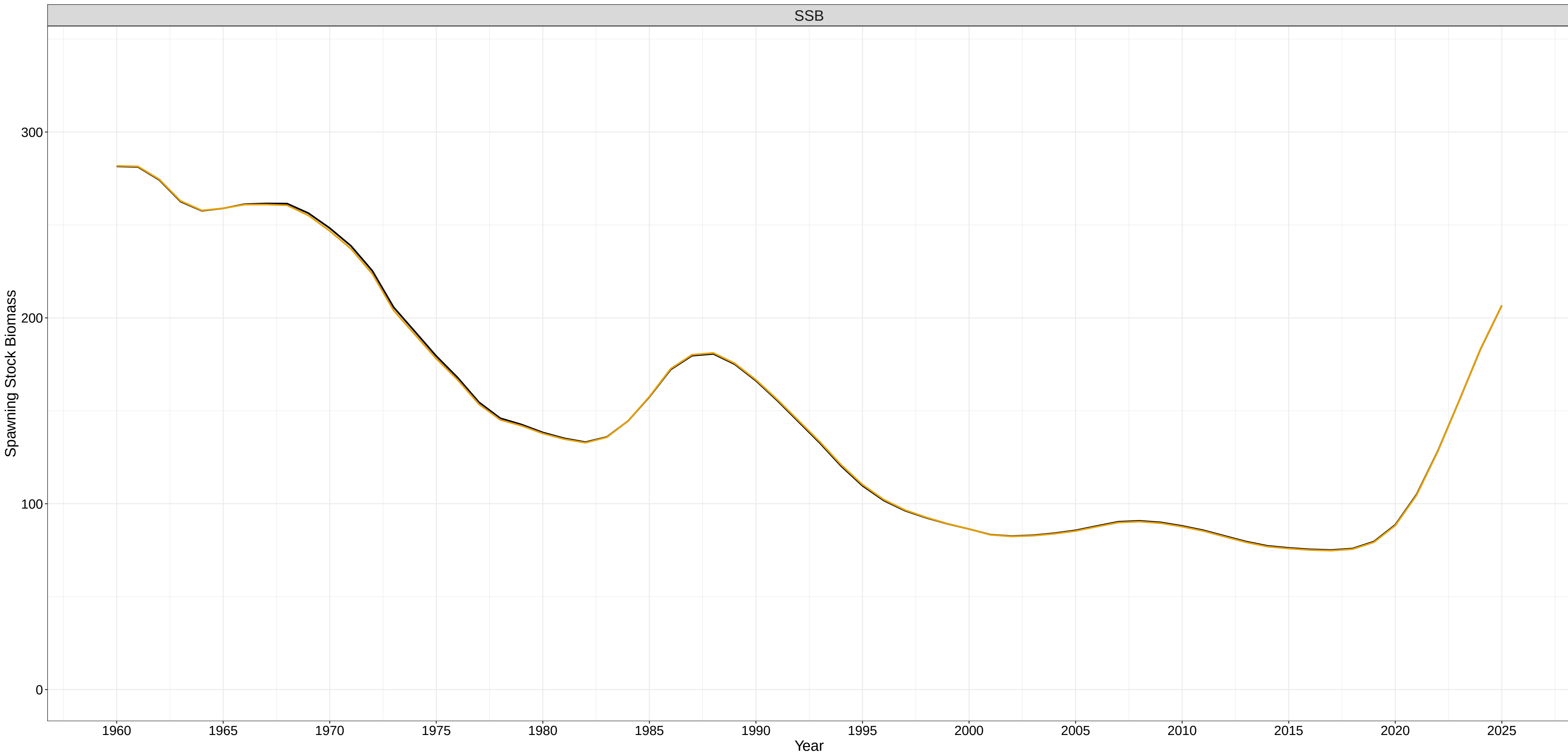


Recruitment

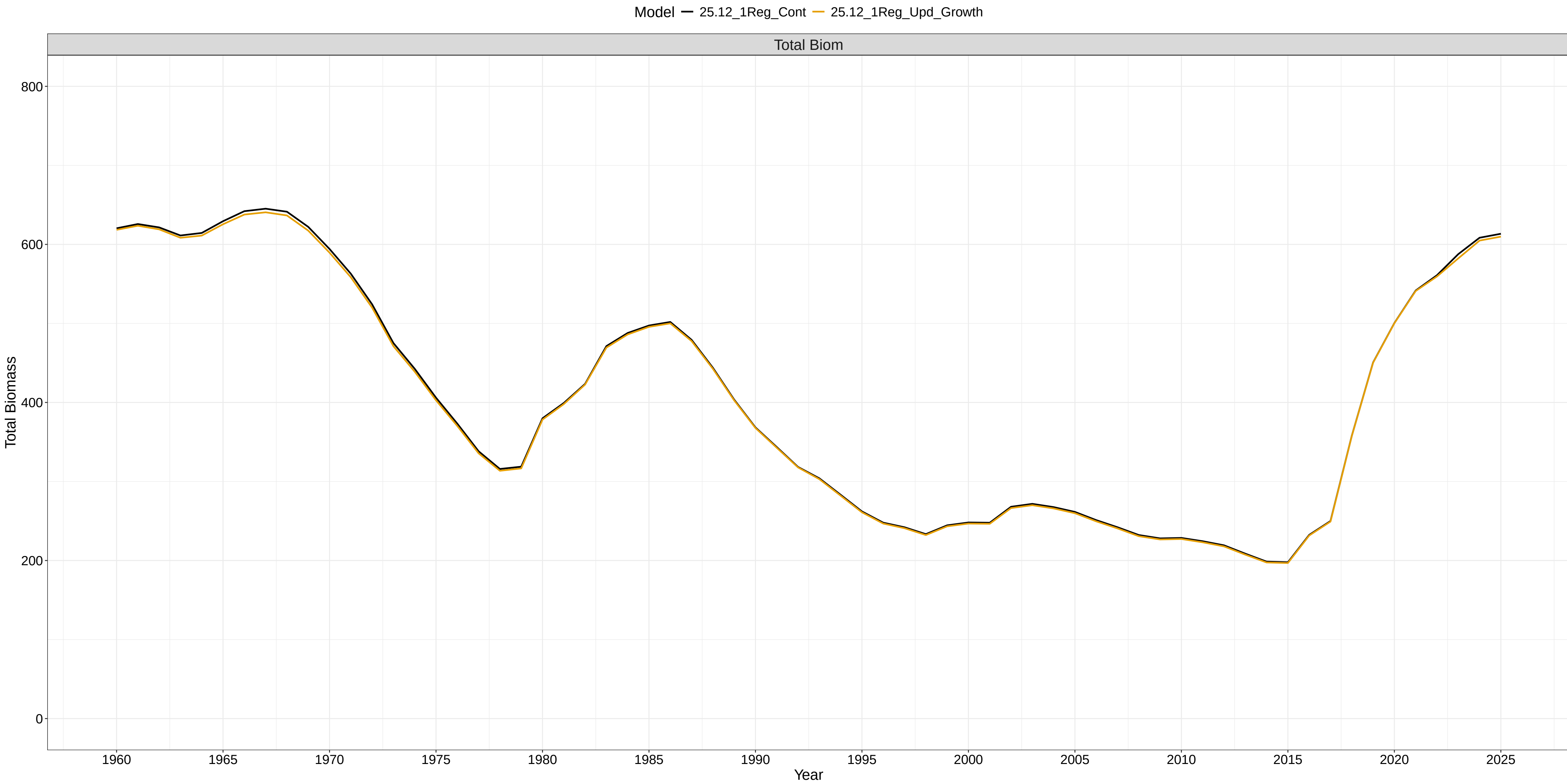


SSB

Model — 25.12_1Reg_Cont — 25.12_1Reg_Upd_Growth

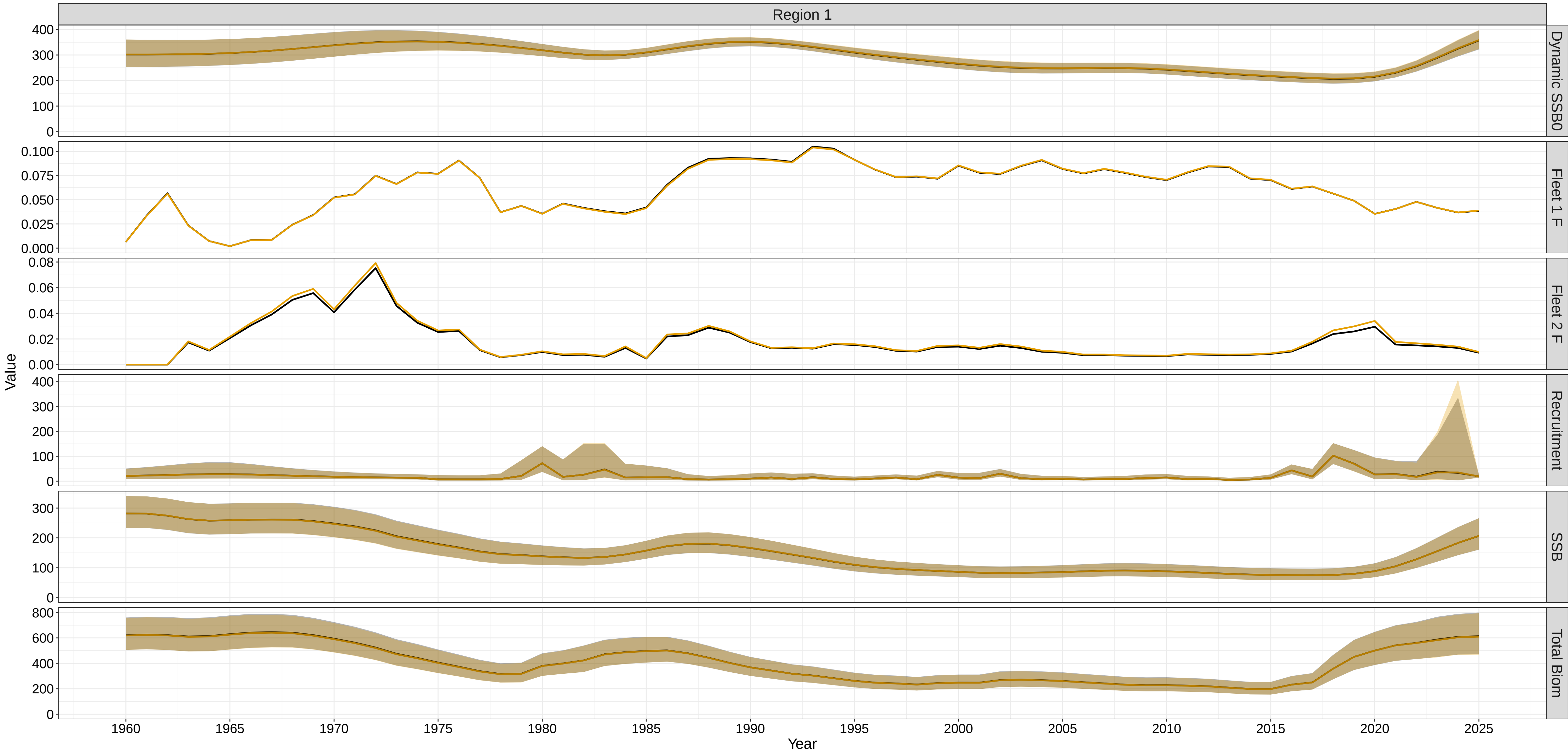


Biomass



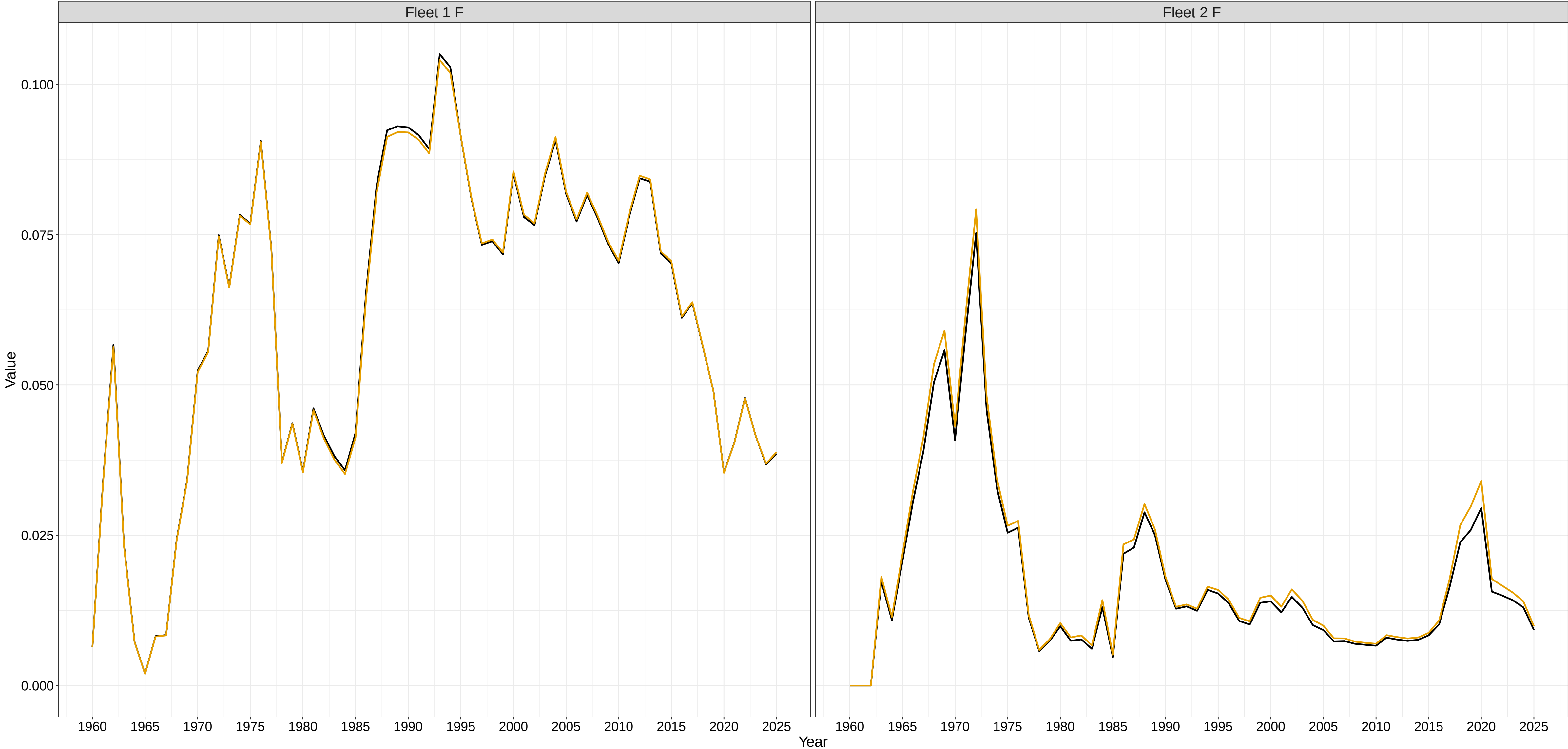
Time Series Plots

Model 25.12_1Reg_Cont 25.12_1Reg_Upd_Growth

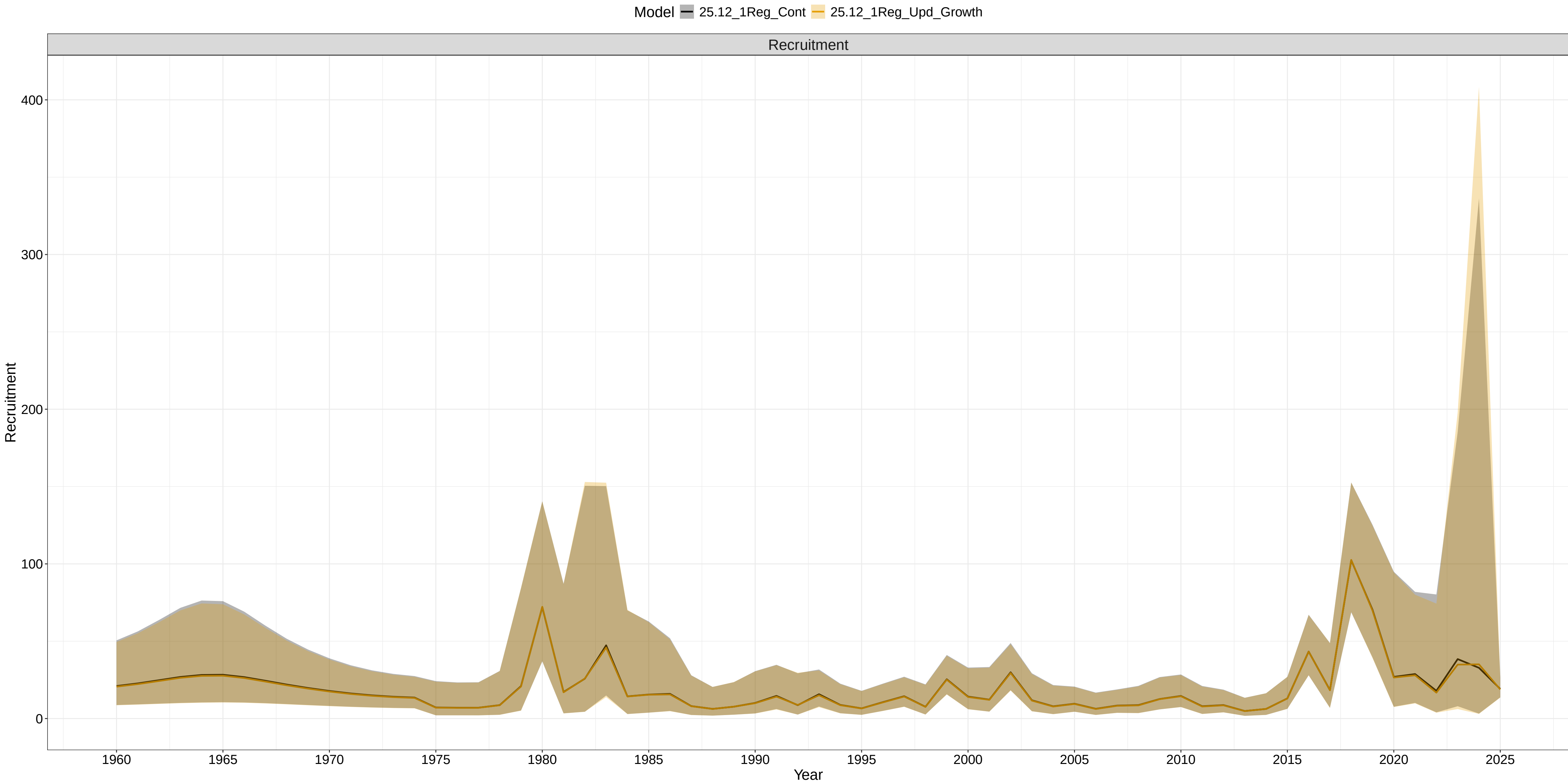


Fishing Mortality

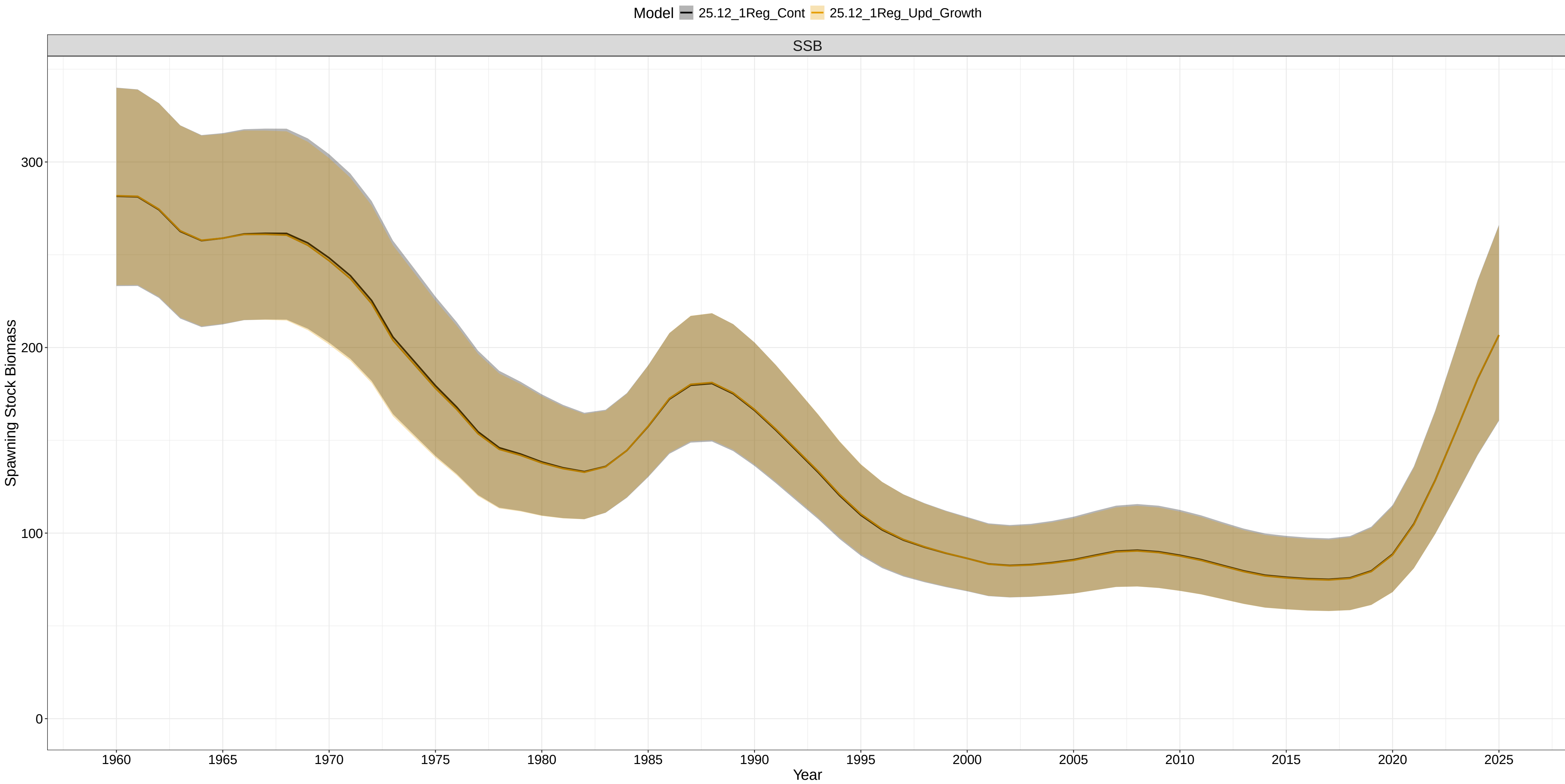
Model — 25.12_1Reg_Cont 25.12_1Reg_Upd_Growth



Recruitment



SSB



Biomass

